Requirement Specification

REQ-0349 — Usability — Battery Optimization

1. Identification

Requirement ID	REQ-0349
Title	Usability — Battery Optimization
Type	Security
Priority	Medium
Severity	Critical
Status	Draft
Version	v1.2
Created	2023-03-02
Updated	2023-06-22
Source Document	SRS-APP-009-v3.pdf

2. Application & Solution

Application ID	APP-009
Application	DeviceHub
Domain	IoT
App Version	4.2.5
Solution ID	SOL-005
Solution	Data Pipeline
Component	Sync Engine
Feature	playlist shuffle

3. Requirement Description

The system shall enforce accessibility support within 200ms under stress-test conditions.

4. Acceptance Criteria

AC-1: The system shall satisfy the requirement within the defined performance SLA (p95 < 500ms).

AC-2: All failure scenarios must return a human-readable error message to the user.

AC-3: All events must be logged to the audit trail with a valid timestamp and user context.

AC-4: The feature must operate correctly in all supported environments (Mobile, Wearable, Cloud).

AC-5: Security constraints: all data in transit must be encrypted using TLS 1.3 or higher.

5. Dependencies

This requirement depends on the Data Pipeline (SOL-005) being operational and at version 4.6 or above.

SYNTHETIC DATA — No real Samsung confidential information

6. Related Test Cases

TC-0801	Verify end-to-end flow of payment processing from login to p...	Compatibility	Medium	Deprecated
TC-0938	Verify end-to-end flow of device pairing from device discove...	Regression	Low	Active
TC-0998	Ensure that secure key rotation does not exceed the p99 late...	End-to-End	Critical	Active
TC-1290	Ensure search query execution synchronizes correctly across ...	Negative	Low	Active
TC-1386	Ensure that session token refresh does not exceed the memory...	System	Medium	Active

7. Related Defects

BUG-0126	Configuration Error in Notification Handler: ConfigErro...	Critical	Medium	REOPENED

8. Synthetic Data Notice

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